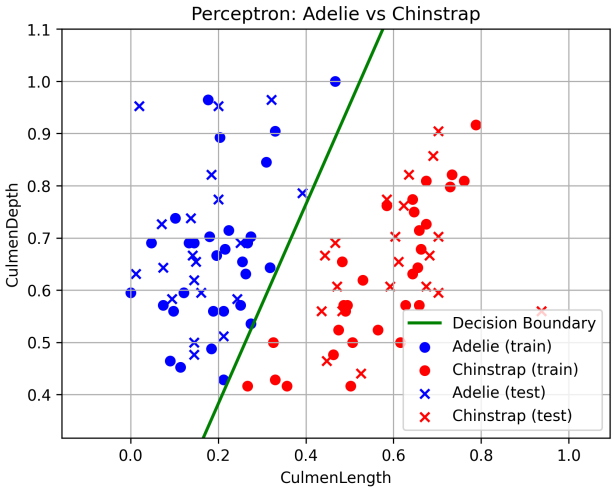
Task 1: Single Layer Perceptron - Visualization & Analysis Report

Student: Anas Hesham | **Course:** Neural Networks | **Dataset:** Penguins (150 samples, 3 species)

Setup

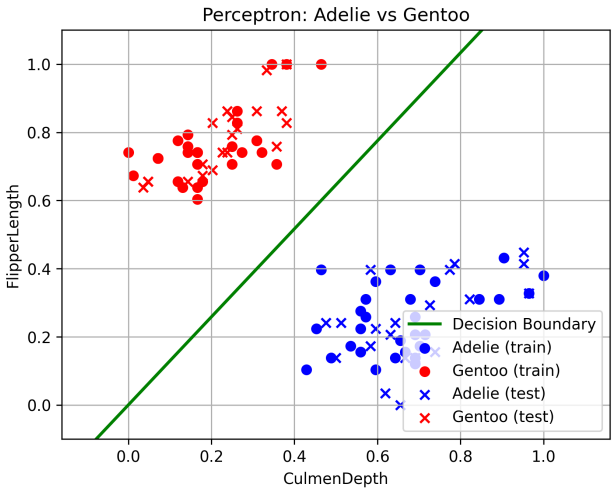
Data cleaning: uses the team's *data_cleaning.py* (output *Ready_penguins.csv*). Missing values are filled with the most frequent value of the same species, so **no row is dropped** (all 150 rows are kept), and numeric features are min-max scaled to [0, 1]. **OriginLocation:** one-hot encoded in data_cleaning.py; the Perceptron combines the three columns into one numeric feature (Torgersen = 0, Biscoe = 0.5, Dream = 1) so it can be used as a single feature. **Training:** 30 random training + 20 test samples per class, learning rate = 0.01, epochs = 100, bias enabled, signum activation, update rule w = w + eta * (t - y) * x. **Evaluation:** the confusion matrix and accuracy are computed manually by counting TP, TN, FP and FN (no scikit-learn). All runs use the same random seed (42); a different random split can change the numbers slightly.

Visualizations (circles = training samples, crosses = test samples, green line = decision boundary)



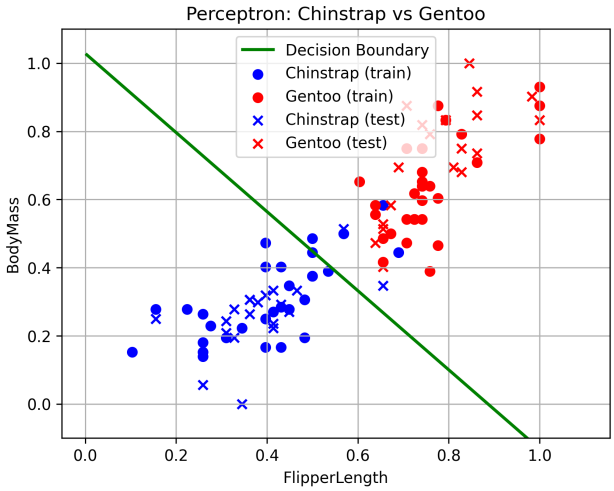
Combination 1: Adelie vs Chinstrap	**Accuracy: 100.0%**
Features: **CulmenLength** + **CulmenDepth**	Result: **Excellent**
Confusion matrix (test): TP = 20, FN = 0, FP = 0, TN = 20	w = [-0.0045, 0.0024], b = 0.00

Analysis: CulmenLength almost splits the two species on its own: Adelie points (short beaks) are on the left and Chinstrap points (long beaks) are on the right, with a clear gap between them. CulmenDepth adds little because both species have similar beak depths, which is why the learned line is steep. All 40 test samples were classified correctly.



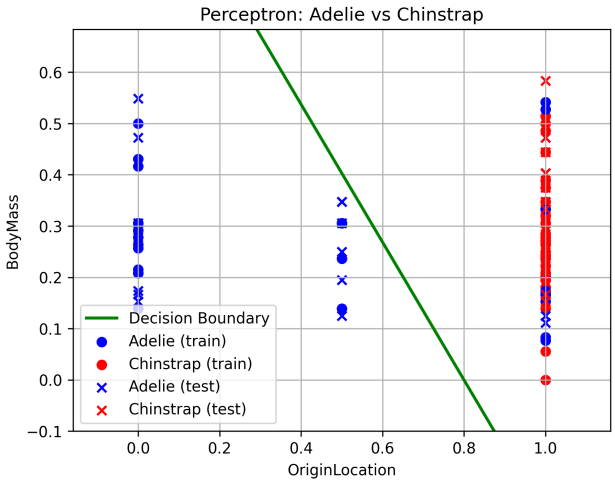
Combination 2: Adelie vs Gentoo	**Accuracy: 100.0%**
Features: **CulmenDepth** + **FlipperLength**	Result: **Excellent**
Confusion matrix (test): TP = 20, FN = 0, FP = 0, TN = 20	w = [0.0138, -0.0107], b = 0.00

Analysis: The two species sit in opposite corners: Gentoo has long flippers and shallow beaks (top-left), Adelie has short flippers and deep beaks (bottom-right). The gap between the clusters is very wide, so the classes are clearly linearly separable and the Perceptron classifies every test sample correctly.



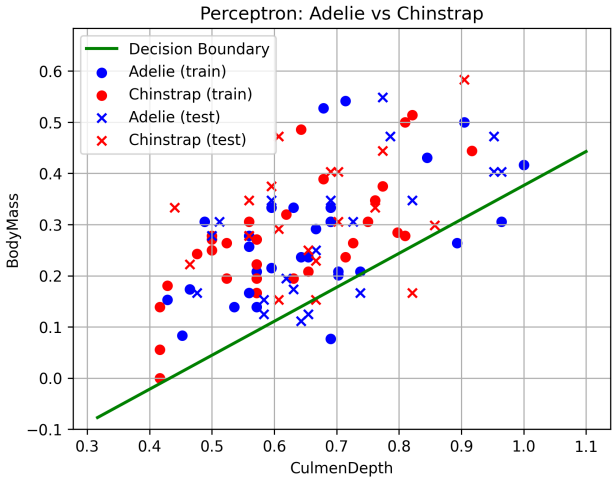
Combination 3: Chinstrap vs Gentoo	**Accuracy: 95.0%**
Features: **FlipperLength** + **BodyMass**	Result: **Good**
Confusion matrix (test): TP = 18, FN = 2, FP = 0, TN = 20	w = [-0.2031, -0.1750], b = 0.18

Analysis: Gentoo penguins are heavier and have longer flippers, so the two classes lie on opposite sides of a diagonal line. However, a few large Chinstraps (FlipperLength around 0.6-0.7) fall inside the Gentoo cluster. They cause the 2 test errors (Chinstrap predicted as Gentoo). These features discriminate well, but not perfectly.



Combination 4: Adelie vs Chinstrap	**Accuracy: 82.5%**
Features: **OriginLocation** + **BodyMass**	Result: **Moderate**
Confusion matrix (test): TP = 13, FN = 7, FP = 0, TN = 20	w = [-0.0500, -0.0372], b = 0.04

Analysis: After preprocessing, OriginLocation has only three values (Torgersen = 0, Biscoe = 0.5, Dream = 1), so the points form three vertical stripes. Torgersen and Biscoe contain only Adelie, but Dream contains every Chinstrap mixed with Adelies of the same body mass. The boundary labels all of Dream as Chinstrap: Adelies from the other islands are correct, while the 7 Adelie test samples from Dream are misclassified.



Combination 5: Adelie vs Chinstrap	**Accuracy: 52.5%**
Features: **CulmenDepth** + **BodyMass**	Result: **Bad**
Confusion matrix (test): TP = 3, FN = 17, FP = 2, TN = 18	w = [0.0462, -0.0696], b = -0.02

Analysis: Adelie and Chinstrap have almost the same body mass and beak depth, so their points are completely mixed and no straight line can separate them. On such data the Perceptron never converges: the weights keep changing and the final line is simply where training stopped after 100 epochs. Most samples ended on the Chinstrap side (3/20 Adelie vs 18/20 Chinstrap correct), giving 52.5%, close to random guessing (50%).

Test accuracy for every feature pair and class pair (same settings)

Feature pair	Adelie vs Chinstrap	Adelie vs Gentoo	Chinstrap vs Gentoo
CulmenLength + CulmenDepth	**100.0%**	**100.0%**	**100.0%**
CulmenLength + FlipperLength	95.0%	100.0%	97.5%
CulmenLength + OriginLocation	95.0%	95.0%	100.0%
CulmenLength + BodyMass	95.0%	95.0%	100.0%
CulmenDepth + FlipperLength	77.5%	100.0%	100.0%
CulmenDepth + OriginLocation	82.5%	95.0%	100.0%
CulmenDepth + BodyMass	52.5%	100.0%	100.0%
FlipperLength + OriginLocation	82.5%	100.0%	100.0%
FlipperLength + BodyMass	70.0%	100.0%	95.0%
OriginLocation + BodyMass	82.5%	87.5%	100.0%

Conclusion - features with the highest accuracy

• **CulmenLength + CulmenDepth** achieved the highest accuracy: **100% for all three class pairs**. It is the only feature pair that does this (highlighted row).

• **CulmenLength is the key feature for Adelie vs Chinstrap**, the hardest pair: every feature pair that includes it reached at least 95.0%, while every pair without it stayed at 82.5% or below.

• Gentoo is easy to separate from both other species: pairs built from CulmenDepth, FlipperLength and BodyMass reach 95-100%, because Gentoos are larger with shallower beaks.

• The lowest accuracy was **CulmenDepth + BodyMass** for Adelie vs Chinstrap (52.5%), where the classes overlap completely.